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LATEST NEWS

REVIEW OF THE NATIONAL CONFERENCE ON MAINTENANCE 4.0: PdM-PHM'26!

On May 13 and 14, 2026, the University of Oran 2 Mohamed Ben Ahmed vibrated to the rhythm of technological innovation during the National Conference on Maintenance 4.0: PdM-PHM'26. Jointly organized by the Laboratory of Industrial Production Engineering and Maintenance (LGPMI) and the Institute of Maintenance and Industrial Safety (IMSI), this event represented a real logistical challenge.

Decided at the last minute, the organization of this edition nevertheless turned out to be a total success, both in terms of the quality of the exchanges and its impact.

Held under the theme of predictive maintenance (Predictive Maintenance - PdM) and systems health management (Prognostics and Health Management - PHM) in the context of Industry 4.0, the conference brought together many doctoral students, researchers and professionals.

Although specifically centered on predictive maintenance, this edition shone through its interdisciplinarity. It brought together researchers from various fields (telecommunications, mechanical engineering, electrical engineering, quality engineering, etc.), all united around the cross-cutting theme of maintenance. This diversity made it possible to broaden perspectives and provide a global and comprehensive vision of the latest scientific advances.

AN UNPRECEDENTED SYNERGY BETWEEN UNIVERSITY AND INDUSTRY

The great particularity of this edition lies in the significant participation of major industrial players, notably Knauf, PLM-Ressources and Sonatrach. For the first time at the national level in this type of event, a dedicated Industrial Session was set up.

Representatives of the socio-economic sector presented concrete problems encountered in the field, directly soliciting the expertise of the scientific community to find innovative solutions. This pioneering initiative made it possible to forge unbreakable links between the academic world and professional practice.

HIGH-LEVEL PLENARY SESSIONS

The scientific program was also honored by the presence of major figures in national research. Participants were thus able to attend two highly rich plenary sessions, led by recognized experts:

- **Prof. Bendiabdellah Azeddine**, who shared his in-depth expertise on the theme: "Electric machines & drives maintenance and fault diagnosis".
- **Prof. Tawfik Benabdallah**, who spoke on experience sharing through his presentation: "Applied Research through National Research Projects experience".

These two masterful and extremely instructive presentations gave rise to dense and passionate debates. The presence of these eminent researchers not only raised the scientific level of the conference, but also enriched the discussions around the oral presentations of the various participants.

Four main thematic axes structured the sessions over these two days:

1. Data and Signal Processing.
2. Fault Detection and Diagnostics.
3. Prognostic - Remaining Useful Life.
4. Decision-Support.

Supported by a high-ranking scientific committee, this edition left a lasting impression. A big thank you to all the participants, as well as our warm congratulations to the scientific and organizing committees for having brilliantly met this challenge and contributed, once again, to positioning our institute at the heart of the industrial and scientific challenges of tomorrow.



The exchange would like to extend sincere gratitude to Dr. Bouchaala M. Abdelghani, Co-chair of the conference, who generously shared this comprehensive review with us.

MASTER FINAL PROJECT DEFENCES - JUNE 2026

The month of June will be marked by the final project defences for the Master's degree, where our second-year Master's students will present their work before a jury composed of academic researchers and professionals from the industry. We wish them every success and warmly thank the supervisors and jury members for their dedication.



I. Saidi
2nd year industrial engineering student

BRAIN ROT: THE SILENT EFFECT OF SOCIAL MEDIA ADDICTION

Imagine that the biggest threat to your brain today is not an illness or an accident. It is something you hold in your hands every day: your phone. Fast browsing, endless scrolling through reels, and constant notifications can slowly change the way your brain works.

Recently, the term "brain rot" has become increasingly popular. It is often linked to excessive phone use and the overconsumption of short-form content on social media. This raises an important question: **Are you in control of your mind, or have social media algorithms started controlling you?**

Brain rot is not an actual disease. Rather, it is the loss of the ability to focus and think deeply due to constant exposure to fast and highly stimulating content. As a result, your brain becomes accustomed to instant entertainment and finds it difficult to engage in activities that require patience and concentration.

Scientifically, this is related to changes in the brain's reward system, especially dopamine. Every time we watch a short video that we enjoy, our brain releases a small amount of dopamine. The more content we consume, the more dopamine we receive.

Over time, the brain adapts to these high levels of stimulation. Eventually, the same amount of dopamine is no longer enough, so we need more scrolling and more content to feel satisfied. This is where addiction begins. The result is increased distraction, reduced focus, and a strong sense of boredom when performing simple daily activities.

The problem is not the screens themselves but the type of content being consumed. In simple terms, there is an entire economy built around capturing and keeping people's attention. This is why many platforms are designed to be addictive. Endless scrolling, unpredictable content, and sudden notifications all contribute to compulsive behavior, making it difficult for users to stop.

Social media algorithms also do not always show us content we enjoy. Sometimes they expose us to shocking, controversial, or even infuriating content. The reason is simple: strong emotions increase engagement, and greater engagement leads to higher profits for the platforms.

Furthermore, short-form content trains our brains to consume information rapidly and become impatient. Over time, this can affect a person's ability to live a balanced and productive life.

Now that we understand what brain rot is and why it is a serious issue, the next question is: **What is the solution?**

Let's look at a study conducted by researchers at the University of California. They took a group of

students on a camping trip and encouraged them to participate in both individual and group activities, with one important condition: no screens were allowed. After just five days, researchers noticed a significant improvement in the students' emotional and social intelligence. This finding suggests that the negative effects of excessive social media use can be reduced and even reversed.

Here is a simple plan to overcome brain rot:

1. **Train the algorithm according to your new interests.**

Start watching meaningful and educational content. On platforms such as YouTube, engage with videos that add value to your life. When using reels or short videos, select "Not Interested" whenever you encounter content that does not benefit you.

2. **Get your brain used to natural sources of dopamine.**

Activities such as sports, reading, hobbies, and studying provide healthier forms of satisfaction. At first, these activities may feel boring compared to social media, but patience is key.

3. **Restore your brain's abilities.**

Practice simple mental exercises such as solving problems, doing calculations, writing, and critical thinking. These activities strengthen focus and cognitive performance.

4. **Improve your environment.**

Keep your phone away while studying or working. You can also use apps that block social media during specific periods to help reduce distractions.

In the end, the human brain is an incredible gift. It should be protected, developed, and used to build a better version of ourselves rather than being consumed by endless streams of fast content.



H. Hadjail

1st year industrial engineering student

IF YOU WAIT UNTIL YOU FEEL MOTIVATED... WILL YOU EVER START?

Imagine this situation: your exam is only one week away. Your lessons have piled up, your desk is clean,

your coffee is ready, and you even have free time. Yet, you still do not feel that urge to get up and sit at your desk to study.



That feeling convinces your mind that there is still plenty of time left and distracts you from thinking about the consequences of delaying your studies. You tell yourself:

I will wait until I am in the right mood to study properly.

But that mood never comes. And before you realize it, you are stuck in the same cycle of procrastination again and again.

Does this sound familiar? Have you ever experienced something like this?

This is one of the most common struggles among students. The problem is usually not a lack of ability, but the absence of that mysterious feeling we call **motivation**.

The problem begins when motivation becomes your alarm clock: if it rings, you study; if it does not, you keep sleeping

In this article, I want to discuss a very common idea: the belief that discipline cannot exist without motivation.

Understanding the difference between these two forces is not just another piece of information. It is one of the most important lessons a student can learn. If taken seriously, it can make a real difference in daily life and help build a more balanced, focused, and fulfilling future with fewer regrets and less frustration.

WHAT IS MOTIVATION AND WHY CAN'T WE RELY ON IT?

If we ask any student what they need to start studying, the answer is almost always the same: **I just need to feel motivated.**

But what exactly is motivation? And why does it disappear at the moments when we need it the most?

From a psychological point, motivation is not just a feeling. It is a complex emotional and neurological process. When we expect to achieve something valuable and meaningful to us, our brain releases a remarkable chemical called **dopamine**. This chemical creates that familiar feeling of excitement, energy, enthusiasm and drive.

However, dopamine is released in response to **expectation**, not to the actual work itself. In other words, we can feel happy and excited about something we have only imagined achieving, even before taking the first step. And once that sense of novelty fades, the dopamine fades with it, and so does the feeling of motivation.

Psychologists *Edward Deci* and *Richard Ryan*, through their well-known **Self-Determination Theory**, explained that human motivation depends on three basic psychological needs:

1. **Autonomy:** feeling that you are in control of your choices.
2. **Competence:** feeling capable of succeeding.
3. **Relatedness:** feeling connected to other people.

When these needs are satisfied, motivation becomes stronger. But when they are missing - which often happens to us as students, especially during our stressful periods - motivation becomes difficult to feel.

And this is where the trap begins: we treat motivation as the starting line, as if without it we simply cannot begin. Waiting for motivation means giving control of our productivity to something we do not fully control, something that can disappear exactly when we need it most.

Just think about the beginning of every academic year. Most of us feel genuinely excited: new notebooks, organized schedules, and big plans. **This year will be different.** We say it and we truly believe it, convinced that this time, things will actually change.

But a few weeks later, after the excitement of the beginning fades, that enthusiasm quietly disappears, and slowly we find ourselves back to the same old habits!

What changed? Not the goal. Not our abilities. Not our potential. The only thing that changed was the **feeling**

WHAT IS DISCIPLINE AND WHERE DOES IT COME FROM?

If motivation is a feeling, then discipline is a decision.

It is not a talent, and it is not something that only a few people are born with. It is a decision that anyone can make at any moment, no matter how they feel.

We often hear phrases like: **He is naturally disciplined** or **I am not just that kind of student** or **I cannot handle it.**

But research in behavioral psychology tells us something different. Discipline is not a superpower that some students have and others do not. It is a skill that is built through repetition, just like a muscle. The more you train it, the stronger it becomes.

So, what does discipline look like in real life?

Psychologist *Angela Duckworth*, who is well known for her research on **grit and perseverance**, found that long-term success does not depend on talent or temporary bursts of motivation. Instead, it depends on the ability to keep showing up and doing the work, especially when the work is difficult or boring.

In her studies, she found that the students who performed best were not necessarily the smartest or the most motivated. They were simply the most consistent and persistent.

There is also an important neurological side to this. Every time we choose to act even when we do not feel like it, we strengthen the **prefrontal cortex**, the part of the brain responsible for decision-making, self-control, and long-term thinking.

In other words, discipline is not just a habit that we build. It is a process that gradually reshapes our brains over time. But perhaps the most important thing to understand about discipline is this: **We do**

not need to feel ready before we start. All discipline needs is for us to take the first step. So do not wait until you feel disciplined. **Start first, and discipline will follow.**

MOTIVATION AND DISCIPLINE, ENEMIES OR ALLIES?

The question we may ask ourselves now is: **So, should we completely ignore motivation?** Of course not. At the end, we are human beings, and emotions are part of who we are.

The idea is not to get rid of motivation, but to stop depending on it!

Most of the time, we study when we feel like it, and stop when we don't. And honestly, that is not a learning system! It is just an emotional reaction. Discipline gives us what motivation never can: **consistency**. And consistency is what actually produces real results over time. So instead of asking yourself: **Do I feel motivated today?** Ask yourself: **Have I decided to study today?**. The first question puts you under the control of your feelings. The second puts you in control.

Get this very well in your mind: if you have a dream, discipline is what will help you achieve it. Motivation can give you a push at the beginning, but it cannot carry you all the way.

- If you rely only on motivation, you may spend your life dreaming.
- If you rely on discipline, you give yourself a chance to turn those dreams into reality.

In the end, I hope you found this article helpful and that it encourages you to start your own journey of change and self-improvement.

I would also like to invite you to remember our fellow student *Mohamed Kader*, who recently passed away after battling cancer. May Allah have mercy on his soul.



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student

WHEN STARS DIE, BLACK HOLES ARE BORN: THE COSMIC LIFE CYCLE EXPLAINED

Can a star that lights up the sky for thousands or even millions of years end its life in complete disappearance into a point that cannot be seen? This question opens the door to understanding complex cosmic phenomena such as stars and black holes.

The aim of this article is to explain the life cycle of stars and the relationship that causes a bright star that illuminates the sky to eventually transform, at the end of its journey, into a black hole that absorbs everything that comes near it.

Stars, according to the definition of the European Space Agency (ESA), are spheres of plasma held together by gravity. They derive their energy from nuclear reactions occurring in their core, and this energy is observed in the form of light and heat.

The life of a star goes through several stages, beginning with the **nebula stage**, which is considered the birthplace of stars. A nebula is a large cloud of gas and dust spread throughout space, mainly composed of hydrogen in large amounts, along with

smaller quantities of helium and other elements. These clouds are relatively cold (2500°C), and no nuclear reactions occur at this stage due to the lack of sufficient energy conditions. However, when an external disturbance occurs, such as the explosion of a nearby star, the material inside the nebula begins to contract gradually under the effect of gravity, leading to the next stage. At this stage, what is known as a *protostar* is formed, where matter gathers into a hot and dense spherical mass. The density at the core increases progressively, and particles continuously collide with each other, converting kinetic energy into heat, causing the temperature to rise significantly.

Nevertheless, nuclear fusion has not yet begun at this stage. The temperature continues to increase until it reaches the conditions required to initiate nuclear reactions, approximately 10 to 15 million degrees Celsius, along with extremely high pressure caused by the gravity of the outer layers of the star. At this point, nuclear fusion begins, where hydrogen nuclei fuse to form helium, releasing enormous amounts of energy that become the star's source of light and heat.

The star reaches a state of equilibrium when its contraction stops, because the outward pressure produced by nuclear reactions inside it becomes equal to the gravitational force pulling it inward. This state is known in astronomy as **hydrostatic equilibrium**. In simple terms, it can be understood as follows: Gravity tries to compress the star inward, while the energy produced by nuclear fusion pushes matter outward. When these two forces are balanced, the star neither collapses nor expands significantly, and it enters a stable state that can last for a long period of time.



After these stages, the life of a star does not end. Instead, it enters a critical phase in which its final fate is determined - whether it will remain as a stellar remnant or transform into a black hole. This outcome depends primarily on the star's mass. In the case of low and intermediate mass stars, once the hydrogen in the star's core is exhausted, it undergoes a *helium flash*, during which it releases energy and increases its temperature. This process leads to the formation of a **white dwarf**, which is a dense, hot stellar remnant. The white dwarf continues to cool and eventually fades away, leaving behind a cold, dark remnant known as a *black dwarf* (theoretical). Unlike massive stars, white dwarfs do not turn into neutron stars or black holes. When the core is exhausted, nuclear fusion in the center stops, leading to the loss of hydrostatic equilibrium discussed earlier. As a result, the core begins to contract under the influence of gravity, while the outer layers expand, causing the star to evolve into a **red giant**. During this stage, helium fusion begins, producing heavier elements such as carbon and oxygen. However, the star does not have sufficient energy to fuse elements heavier than these, so its evolution stops at this point. Over time, the outer layers are expelled into space,

forming a gaseous cloud known as a *planetary nebula*, while the core remains as a small, extremely dense object called a white dwarf. No nuclear fusion occurs in this remnant, and its temperature gradually decreases over billions of years.

In massive stars, temperature and pressure rise significantly due to the strong gravitational force, allowing advanced nuclear fusion reactions in which heavier elements are formed through successive stages: carbon → oxygen → neon → silicon → then oxygen → neon → silicon → iron. When the iron stage is reached, energy production stops because iron fusion does not release energy; instead, it consumes it. As a result, the internal balance of the star collapses. Once energy production ceases, the core collapses extremely rapidly inward, and the density increases dramatically. This is followed by a sudden rebound that leads to a powerful explosion: a **supernova**. This explosion releases enormous amounts of energy and ejects heavy elements into space, contributing to the formation of new elements in the universe. The explosion also generates shock waves that propagate through gas clouds and can trigger their collapse, thereby helping in the formation of new stars.

After a supernova, the fate of the star is determined by the mass remaining after the explosion. If this mass is moderate, a **neutron star** is formed. It is a very small and extremely dense object resulting from the collapse of atoms under enormous pressure, where protons and electrons combine to form neutrons. However, if the remaining mass is extremely large, the collapse continues without stopping because no force is strong enough to counteract it. As the contraction continues, the matter is compressed into an extremely dense point called the *singularity*. Around this point, the *event horizon* is formed - a region from which nothing can escape, not even light. The final result is the formation of a **black hole**, one of the most mysterious objects in the universe.



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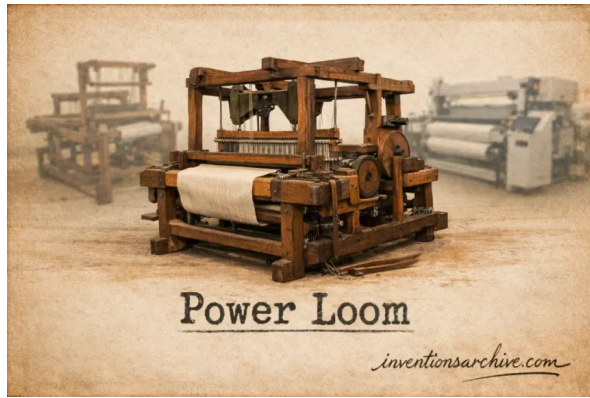
INDUSTRY 1.0: HOW THE FIRST INDUSTRIAL REVOLUTION CHANGED OUR WORLD

Imagine living in the year 1750. There were no factories, no machines, and no engines. People worked using only their muscles. They worked in dark homes to make clothes, or in simple workshops to forge metal. Back then, making clothes was a **Domestic System** (home-based work). Production was very slow. It could not meet the growing market needs. Today, industrial engineers call this problem a **Capacity Shortage**.

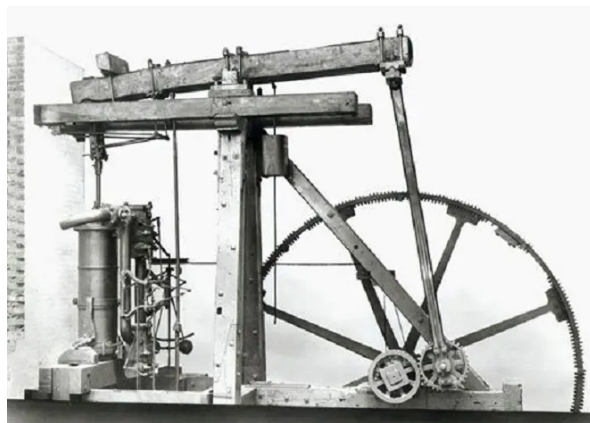
Everything changed with two great inventions. In 1764, James Hargreaves invented the **Spinning Jenny** :



Later, in 1785, Edmund Cartwright invented the **Power Loom** :



These machines automated textile production. Output multiplied by ten times! Today, we call this historic milestone **Process Automation**. However, these early machines needed **Water Power**. Therefore, owners had to build factories only next to rivers. This was a big geographic limit. Then, the Scottish engineer James Watt created a miracle. He redesigned an old steam engine. Watt added a separate condenser and changed linear motion into rotational movement. He created the **Rotary Steam Engine**. This amazing machine converted thermal energy into mechanical energy.



Watt's Rotary Steam Engine manufactured - Bolton 1788 - Birmingham

Watt's invention changed the science of **Thermodynamics** forever. It used much less coal. Early steam engines only pumped water out of deep coal mines because their movement was shaking and unstable. But Watt's engine was smooth and efficient. It started a global **Energy Transition**. Now, factories could operate anywhere in the world!

With these new machines, a new challenge appeared: How do we organize humans to work efficiently around a non-stop machine? This question started **Industrial Management** and created **The Factory System**. This new system had three main pillars:

1. **Centralization:** Machines were too expensive for workers to buy. Therefore, rich owners bought the machines and put them under one big roof: the factory.
2. **Clock Time:** Machines do not sleep. So, factories replaced flexible working hours with strict, controlled schedules.
3. **Division of Labor:** Instead of one artisan making a whole product, factories divided the work into simple, repetitive steps. This created a need for **Unskilled Labor**.

Industry 1.0 changed our planet. It created massive **Iron Foundries** and huge coal mines. However, it also had dark sides. Factories used child labor for low wages. This caused anger, and a group called the Luddites started smashing machines because they feared losing their jobs. This was history's very first fight between **humans and automation!**



I. DROUCHE

2nd year industrial engineering student

THE CURSED NUMBER THAT KEPT POPPING UP IN PHYSICS

« *The shortest path between two truths in the real domain passes through the complex domain.* »

Jacques Hadamard

In 1926, Erwin Schrödinger published his theory of quantum mechanics, introducing a new way to explain particles and their behavior. This theory remains our current best description of the universe. Schrödinger introduced a framework for studying particles, governed by his now-famous equation known as the Schrödinger equation (we omit details):

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(x, t) \right] \psi$$

However, something peculiar lurked within this equation: the letter i , representing the imaginary unit. Many physicists disliked the presence of an imaginary number in a fundamental law of nature, and even Schrödinger himself was uneasy with his own equation. But how did a single number provoke such a backlash? And why did mathematicians choose to call it “imaginary” in the first place?

Thousands of years ago, ancient mathematicians were able to solve various types of equations, yet their methods did not rely on algebra, simply because algebra had not yet been invented. At the time, mathematics was expressed in words and geometric pictures rather than symbolic formulas. Take, for example, the equation $x^2 = 4$. Ancient mathematicians would have interpreted this as the area of a square with side length x being equal to 4. In fact, this is precisely why we call x^2 “ x squared.” Consequently, their methods were heavily rooted in visual geometry. One of their remarkable achievements was solving quadratic equations, those of the form $ax^2 + bx + c = 0$. However, because they relied on geometry, they encountered a clear contradiction. Consider $x^2 + 1 = 0$; to solve this, x^2 must equal -1 . But how can an area be negative? Their solution was simply to ignore such cases, as it did not trouble them—after all, what practical use could a non-existent number have? And while they successfully solved many types of equations, ancient mathematicians struggled greatly with a special class: the cubic equations.

The cubic was such a tough problem that even Luca Pacioli (Leonardo da Vinci's mathematics professor) considered it unsolvable. It was not until 1545 that the mathematician Gerolamo Cardano published a solution. The secret to solving it lay in separating mathematics from the physical world and inventing a new kind of number, one that could represent a negative area, something that had no counterpart in reality. You see, the problem Cardano faced when solving the cubic was that his method reduced the cubic to a quadratic, which could be solved using old geometric techniques, yet some equations yielded a negative value under the square root. To bypass this obstacle, Cardano introduced a new number: $\sqrt{-1}$. By plugging it into the equation, the solution emerged. (This number would later be called the imaginary unit “ i ” by René Descartes, precisely because of its elusive, non-intuitive nature.) This

number could not be processed alongside regular numbers since real numbers were grounded in physical reality, so the solution was to express numbers as a combination of a real part and an imaginary part, in what we now call a complex number.

Over the years, mathematicians continued to study these complex numbers, developing ways to write them in trigonometric form. However, it was Leonhard Euler who discovered an elegant property that made them truly unique—Euler's formula:

$$\cos \theta + i \sin \theta = e^{i\theta}$$

This was groundbreaking. Working with exponentials is often much easier due to their convenient properties. Since any real number can be expressed in complex form, mathematicians could convert expressions to exponentials using Euler's formula, perform their calculations, and then convert back to trigonometric form taking only the real part at the end. This method became a powerful tool for simplifying computations, and physicists soon adopted it as well. For centuries, physicists used this trick regularly in their calculations, always ensuring that the final result came out real—after all, physics describes reality, so the imaginary numbers were treated merely as a convenient intermediate step. Whenever the imaginary unit appeared, they avoided it or eliminated it by the end. But when Schrödinger applied this method, it did not work as usual. He used imaginary numbers purely as a calculational tool, fully expecting to discard them at the end—especially since this was not just any theory; it was meant to describe the very heart of the universe. For Schrödinger and his contemporaries, the final solution simply had to be real. Yet, to their astonishment, it wasn't. The imaginary part stubbornly remained.

The physicist Freeman Dyson later commented: “Schrödinger put the square root of minus one into the equation, and suddenly it made sense. Suddenly it became a wave equation instead of a heat conduction equation. And Schrödinger found to his delight that the equation has solutions corresponding to the quantized orbits in the Bohr model of the atom. It turns out that the Schrödinger equation describes correctly everything we know about the behavior of atoms. It is the basis of all of chemistry and most of physics. And that square root of minus one means that nature works with complex numbers and not with real numbers. This discovery came as a complete surprise to Schrödinger as well as to everybody else.” What began as a quirky intermediate trick to solve cubic equations turned out to be fundamental to the description of reality. Only by giving up mathematics' direct connection to everyday reality could we uncover nature's true underlying structure. From that point on, imaginary numbers kept appearing again and again, consistently proving their presence in our world.

Interestingly, a recent study suggests a new method that may avoid working with complex numbers altogether when deriving the Schrödinger equation. Perhaps, at long last, Schrödinger's dream of seeing only real numbers in his equation may yet come true.

Science is elegant.